

Name _____ Period _____

Skill 20.2 Exercise 1

In the space below, import the pandas library, then create a DataFrame with the following data,

Product ID	Product Name	Color
1	t-shirt	blue
2	t-shirt	green
3	skirt	red

Skill 20.3 Exercise 1

You're running a chain of pita shops. You want to create a DataFrame with information on your different store locations.

In the space below, import the pandas library, then use a list of lists to create a DataFrame with the following data:

Store ID	Location	Number of Employees
1	San Diego	100
2	Los Angeles	120
3	San Francisco	90

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Skill 20.4 Exercise 1

Read the CSV file below into a DataFrame called *df*.

city_data.csv

	City	Population	Median Age
0	Maplewood	100000	40
1	Wayne	350000	33
2	Forrest Hills	300000	35
3	Paramus	400000	55
4	Hackensack	290000	39

Save the DataFrame you created above to a new CSV file called *new_city_data.csv*

Skill 20.5 Exercise 1

Load the CSV *imdb.csv* into a variable called *df*, so that you can learn about popular movies from the past 90 years.

Print the first 5 rows of the dataset

Print the first 7 rows of the dataset

Print the info associated with the dataset

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Skill 20.6 Exercise 1

Refer to the DataFrame below which shows data collected by four health clinics run by the same organization. Each row represents a month from January through June and shows the number of appointments made at four different clinics .

```
df = pd.DataFrame([
    ['January', 100, 100, 23, 100],
    ['February', 51, 45, 145, 45],
    ['March', 81, 96, 65, 96],
    ['April', 80, 80, 54, 180],
    ['May', 51, 54, 54, 154],
    ['June', 112, 109, 79, 129]],
    columns=['month', 'clinic_east',
             'clinic_north', 'clinic_south',
             'clinic_west']
)
```

You want to analyze what's been happening at the North location. Create a variable called *clinic_north* that contains ONLY the data from the column *clinic_north*.

Skill 20.6 Exercise 2

You want to compare visits to the Northern and Southern clinics in the DataFrame from the previous exercise. Create a variable called *clinic_ns* that contains ONLY the data from the columns: *clinic_north* and *clinic_south*.

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Skill 20.7 Exercise 1

Refer to the DataFrame below which shows data collected by four health clinics run by the same organization. Each row represents a month from January through June and shows the number of appointments made at four different clinics .

```
df = pd.DataFrame([
    ['January', 100, 100, 23, 100],
    ['February', 51, 45, 145, 45],
    ['March', 81, 96, 65, 96],
    ['April', 80, 80, 54, 180],
    ['May', 51, 54, 54, 154],
    ['June', 112, 109, 79, 129]],
    columns=['month', 'clinic_east',
             'clinic_north', 'clinic_south',
             'clinic_west']
)
```

You're getting ready to staff the clinic for March this year. Write a command that will produce a Series made up of the March data. Print the results.

One of your doctors thinks that there are more clinic visits in the late Spring. Write a command that will produce a DataFrame made up of the data for April, May, and June for all four sites. Print the results.

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Skill 20.8 Exercise 1

Refer to the DataFrame below which shows data collected by four health clinics run by the same organization. Each row represents a month from January through June and shows the number of appointments made at four different clinics .

```
df = pd.DataFrame([
    ['January', 100, 100, 23, 100],
    ['February', 51, 45, 145, 45],
    ['March', 81, 96, 65, 96],
    ['April', 80, 80, 54, 180],
    ['May', 51, 54, 54, 154],
    ['June', 112, 109, 79, 129]],
    columns=['month', 'clinic_east',
             'clinic_north', 'clinic_south',
             'clinic_west']
)
```

You're going to staff the clinic for January of this year. You want to know how many visits took place in January of last year, to help you prepare. Create variable `january` using a logical statement that selects the row of `df` where the 'month' column is 'January'. Print the results.

You want to see how the number of clinic visits changed between March and April. Create the variable `march_april`, which contains the data from March and April. Do this using two logical statements combined using `|`, which means “or”. Print the results.

Another doctor thinks that you have a lot of clinic visits in the late Winter. Create the variable `jan_feb_mar`, containing the data from January, February, and March. Do this using a single logical statement with the `isin` command. Print the results.

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Skill 20.9 Exercise 1

Code

The code below returns the DataFrame shown,

```
df = pd.DataFrame([
    ['January', 100, 100, 23, 100],
    ['February', 51, 45, 145, 45],
    ['March', 81, 96, 65, 96],
    ['April', 80, 80, 54, 180],
    ['May', 51, 54, 54, 154],
    ['June', 112, 109, 79, 129]],
    columns=['month', 'clinic_east',
            'clinic_north', 'clinic_south',
            'clinic_west']
)
result = df[df['month'].isin(['March', 'May'])]
print(result)
```

Output

	month	clinic_east	clinic_north	clinic_south	clinic_west
2	March	81	96	65	96
4	May	51	54	54	154

Notice the indices are not consecutive. Use the *reset_index* method to reset the indices and print the results.

Add the *drop* and *inplace* keywords to the index method.

What do the *drop* and *inplace* keywords do?